Problem Set 4. Causality and Studies
Quantitative Methods for Humanities

1. Efficacy of Small Class Size in Early Education

The STAR (Student–Teacher Achievement Ratio) Project was a four-year longitudinal study investigating the effects of class size in early grades on educational performance and personal development. This study, conducted from 1985 to 1989, involved 11,601 students who were randomly assigned to one of three groups: small classes, regular-sized classes, or regular-sized classes with a teacher's aid. The experiment cost approximately $12 million, and although the program ended in 1989 when the first kindergarten cohort completed third grade, data collection continued through high school.

The data file, `STAR.csv`, contains the following variables (see Table 1):

Table 1: Description of Variables in the STAR Study

Variable	Description
`race`	Student's race (1 = white, 2 = black, 3 = Asian, 4 = Hispanic, 5 = Native American, 6 = others)
`classtype`	Type of kindergarten class (1 = small, 2 = regular, 3 = regular with aid)
`g4math`	Total scaled score for the math portion of the fourth-grade standardized test
`g4reading`	Total scaled score for the reading portion of the fourth-grade standardized test
`yearssmall`	Number of years spent in small classes
`hsgrad`	High-school graduation status (1 = graduated, 0 = did not graduate)

The dataset contains missing values, which occur due to students leaving STAR schools before third grade or joining later. The tasks below analyze the effects of small class sizes.

1. **Recode Variables:** Create a new factor variable, `kinder`, in the data frame to recode `classtype` into meaningful labels (`small`, `regular`, `regular_aid`). Similarly, recode `race` into four categories (`white`, `black`, `hispanic`, `others`) by combining `Asian` and `Native American` into `others`. For the race variable, overwrite the original variable.

2. **Comparison of Test Scores:** Compare the fourth-grade reading and math test scores between students assigned to small classes and those in regular-sized classes. Use means to make this comparison while removing missing values. Interpret the results. To assess the size of the effect, compare it to the standard deviation of the test scores.

3. **Quantile Treatment Effects:** Instead of focusing solely on mean differences, examine the distribution of test scores. Compare high scores (66th percentile) and low scores (33rd percentile) for small classes and regular classes. Discuss whether this analysis provides additional insights compared to the mean-based comparison.

4. **Effect of Years in Small Classes:** Some students participated in small classes for all four years, while others only experienced one year.

   (a) How many students fall into each category? Create a contingency table of proportions using `kinder` and `yearssmall`.

   (b) Investigate whether the number of years in small classes affects test scores. Compare the mean and median reading and math scores for students with different durations of exposure to small classes.

5. **Reducing Racial Achievement Gaps:** Investigate whether small class sizes reduced racial achievement gaps. Compare the mean reading and math test scores between white and minority students (blacks and Hispanics) within regular-sized classes. Repeat this comparison for small classes. Provide a substantive interpretation.

6. **Long-Term Effects on Graduation:** Assess whether the STAR program influenced high-school graduation rates.

   (a) Compare graduation rates across different class types (`kinder`).

   (b) Investigate whether the number of years spent in small classes affects graduation rates.

   (c) Analyze whether the STAR program reduced racial disparities in graduation rates.

2. The Anchoring Effect

The **anchoring effect** is a cognitive bias in which individuals' decisions are affected by an initial piece of information, called the *anchor*. This phenomenon was first formally identified and studied by Nobel laureate Daniel Kahneman and his coauthor Amos Tversky in their 1974 paper on judgment under uncertainty.[1]

Originally, Kahneman and Tversky asked participants to estimate the percentage of African countries in the United Nations, after presenting them with a rigged "wheel of fortune" that landed on 10 or 65. In a more recent replication effort (ManyLabs 1 study), participants were asked about the height of Mount Everest. Some were randomly given a "high" anchor, others a "low" anchor, and each participant then provided a numerical estimate of Everest's height. The replication data is contained in the `anchoring.RData` dataset, whose relevant variables are described in Table 2.

Table 2: Key Variables in the `anchoring` Dataset

Variable	Description
`anchor`	The randomly assigned anchor (`low` or `high`)
`everest_feet`	Judged height of Mount Everest in feet
`sex`	Participant's reported sex (`m` or `f`)
`age`	Participant's age in years
`citizenship`	country code of citizenship
`lab_or_online`	Indicator of whether the study was done online or in-lab

Answer the following:

[1] Tversky, Amos, and Daniel Kahneman. 1974. Judgment under uncertainty: heuristics and biases. Science 185 (4157): 1124–1131.

1. **Treatment vs. Outcome.** What is the *treatment* in this experiment, and what is the *outcome* variable of interest?

2. **Threats to Internal Validity.** Identify two factors that could undermine internal validity. Think about sample size or potential interference among participants (e.g., if they share information).

3. **Randomization in online tests?** ManyLabs 1 ensured that the experiment is completely randomized. Explain why there are good reasons to be suspicious about this randomization for online takers of the test?

4. **Checking randomization.** What would you expect in the data under randomization? Load the data (`load(anchoring.RData)`) and check it for the subsamples online vs lab.

5. **Causal inference under randomization.** Suppose that randomization holds for the lab sample? How might one estimate the *causal* effect of the anchor?

6. **Data Exploration.** Load the dataset (`load(anchoring.RData)`) into `R`, then compute and compare the average judged heights (`everest_feet`) by `anchor`.

7. **Interpretation.** Is there a noticeable difference between the "low" and "high" anchor groups? Compare it to the standard deviation of `everest_feet`.

8. **Causation?** Based on this design, can you say that the anchor *caused* people to give higher or lower estimates? Why or why not?

9. **External Validity.** Comment on threats to the external validity of this study. Compare the estimated causal effect different countries to illustrate this potential lack of external validity of randomized experiments.

3. Worker Injury Benefits and Time Out of Work

How worker injury benefits affect time out of work is a central question in labor policy. Some argue that generous benefits reduce incentives to return to work, while others see them as promoting fairness and better job matches.

In July 1980, the state of Kentucky raised the maximum weekly worker injury benefit from \$131 to \$217 (**66%** increase). Only high-earning workers (i.e., above a certain income threshold) qualified for the higher benefits, creating a natural experiment for researchers.[2] The file `injury.csv` contains the variables described in Table 3.

Table 3: Description of Variables in the Worker Injury Data

Variable	Description
`afchnge`	Equals 1 if the claim occurs after the policy change, 0 otherwise
`ldurat`	Log of the duration (in weeks) out of work
`highearn`	Binary indicator (1 if high wage earner, 0 if low wage earner)

Answer the following questions:

[2] Meyer, B.D., W.K. Viscusi, and D.L. Durbin. 1995. Workers' Compensation and Injury Duration: Evidence from a Natural Experiment. *The American Economic Review* 85 (3): 322–340.

1. Identify the treatment and the outcome variables.
2. Would a Difference-in-Mean estimator between the treated and not-treated groups *after* the policy fairly represent the causal effect in this study? Explain.
3. Suppose you decide to stick to a Before-and-After (BA) design. What would be the treated and control groups in such a case?
4. Load the data `injury.csv` and show the summary.
5. Compute the BA estimator and comment on your result.
6. Discuss potential limitations of the BA study, and how the Difference-in-Differences (DiD) design could mitigate them.
7. Why do we need the assumption that, in the absence of the policy, time trends for both groups (high earners and low earners) would have moved in parallel?
8. Recall the DiD estimator:

$$\widehat{\text{SATE}}_{\text{DiD}} = (\overline{Y}_T^{\text{after}} - \overline{Y}_T^{\text{before}}) \; - \; (\overline{Y}_C^{\text{after}} - \overline{Y}_C^{\text{before}}).$$

   Explain, in plain words, the two terms in parentheses.
9. Compute the DiD estimator and comment on the results.
10. List potential factors that might invalidate the parallel trends assumption (e.g., macro shocks hitting the groups differently).